

MOHAMED SATHAK ASFAAN

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EDUCATION

Measi Institute of Information Technology

Master of Computer Applications · Chennai Aug 2024 – May 2026

The New College

Bachelor of Computer Applications · Chennai Aug 2021 – Apr 2024

EXPERIENCE

SMAART Healthcare — Software & Data Intern

Aug 2025 – Nov 2025

- Architected a **real-time AI nurse triage assistant** using **Flask**, **MongoDB**, and **Vapi.ai**, reducing manual patient intake **effort by 40%** by automating symptom collection and routing across concurrent clinic sessions.
- Engineered **NLP-based** intent classification pipelines for voice and text interactions using keyword-pattern scoring and entity extraction, improving patient query routing **efficiency by 35%**.
- Built an **automated** urgency-scoring and **nurse alert escalation system** evaluating patient severity in real time via weighted symptom heuristics, accelerating emergency response workflows by **30%**.

PROJECTS

VConnect – AI-Powered Rural Data Analytics Platform | Python, Flask, MongoDB, Pandas, REST APIs, Java

2025

- Designed and developed a scalable analytics platform to process and visualize data for 650,000+ Indian villages, enabling efficient exploration of demographic, healthcare, education, and infrastructure datasets.
- Built optimized **Pandas** data processing pipelines for cleaning, transforming, and aggregating large CSV datasets, reducing preprocessing time by 50% through vectorized operations and chunked bulk loading.
- Developed **RESTful APIs using Flask and Spring Boot** to connect frontend modules with backend services, enabling secure and efficient access to village statistics, government schemes, and analytics data.
- Implemented AI-powered search and intelligent filtering using full-text indexes and re-ranking to retrieve village-level insights from user queries, improving data retrieval efficiency by 45%.

AI Nurse Triage Voice Assistant | Python, Flask, Vapi.ai, MongoDB, NLP

2025

- Developed a real-time AI voice assistant integrating **Vapi.ai**, **STT/TTS** with a Flask backend to collect patient symptoms through natural conversation before doctor consultation, reducing **intake effort by 40%**.
- Built a multi-stage **NLP pipeline** for symptom extraction, intent classification, and urgency assessment; designed emergency alert workflows identifying critical symptoms and notifying healthcare staff immediately.
- Developed **Flask REST APIs** to manage patient sessions, conversation history, and AI-generated summaries; stored transcripts securely in MongoDB with automatic conversation summarization for clinical reference.

AI Powered Personal Portfolio Website | HTML, CSS, JavaScript, Flask, Python, RAG

2026

- Designed and developed a responsive **portfolio website** and built an **AI assistant** using **Retrieval-Augmented Generation (RAG)** to answer visitor queries from project documentation and resume data with high accuracy.
- Architected a **three-tier AI fallback system**: LLM API → RAG knowledge base → rule-based **chatbot**, ensuring uninterrupted response availability; developed **Flask REST APIs** for semantic query retrieval and context-aware responses.
- Implemented responsive UI with smooth animations and interactive showcases using HTML, CSS, and JavaScript; optimized frontend performance via modular JS, lazy loading, and efficient API communication.

LogiSense – Shipment Tracking & Analytics System | Python, Flask, MongoDB, REST APIs, JavaScript

2026

- Engineered a modular logistics platform with scalable **Flask RESTful APIs** for shipment lifecycle management, real-time tracking, warehouse operations, and automated delay notifications to improve operational visibility.
- Built analytics dashboards displaying fleet utilization, delivery delays, and operational KPIs; optimized MongoDB queries and indexing strategies to support large-scale logistics operations with improved response times.
- Applied **OOP principles** and layered architecture (Controller → Service → Repository) with automated shipment monitoring workflows, reducing manual tracking effort and improving operational reliability.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML, CSS

Frameworks & APIs: Flask, Spring Boot, REST APIs, Pandas, RAG

AI / Machine Learning: Machine Learning, Retrieval-Augmented Generation (RAG), NLP, Intent Classification, Entity Extraction, Voice AI (Vapi.ai), Pandas

Databases: MongoDB, MySQL, **Pinecone** (Vector Database)

Tools & Core CS: Git, **GitHub**, VS Code, Jupyter Notebook, DSA, OOP, DBMS, Docker, Eclipse IDE

ACHIEVEMENTS

Measi Institute of Information Technology — 1st Place — Intra-College Hackathon (Analytics Project)

2025

MGR University – Dept. of IT — 2nd Prize — Inter-College Hackathon Competition

2025

LeetCode — **150+ DSA** problems solved — Medium/Hard focus on algorithms & problem-solving

Ongoing